

Neural Waypoint-to-Control Generation: WN&C Nets

What the method is

The local paper *Imitation Learning-Based Online Time-Optimal Control with Multiple-Waypoint Constraints for Quadrotors* trains waypoint-constrained navigation and control networks (WN&C Nets) on state-control pairs produced offline by complementary-progress-constraint (CPC) time-optimal optimization. Its input contains relative positions to two waypoints, velocity, roll-pitch-yaw attitude, and thrust. Its output is the next angular-rate command and total-thrust command. Delay-aware training maps a delayed observed state to the future command.

Why it can reduce amplification

The deployed network directly predicts control at each update. It therefore does not obtain feed-forward control by repeatedly differentiating a fitted seventh-order position polynomial. This can remove polynomial conditioning, high derivative peaks, segment-duration sensitivity, and wall-clock phase amplification from the waypoint-to-control conversion. Smoothness is learned from CPC demonstrations rather than guaranteed analytically, so prediction errors, distribution shift, delay mismatch, and control discontinuities can still increase s_u or s_x .

Compatibility and limitations

The paper uses full quadrotor state and thrust/angular-rate commands, whereas the current repository calibrates a planar damped double integrator with $x = [p_x, p_y, v_x, v_y]$ and $u = [u_x, u_y]$. A compatible adoption therefore requires a separately generated expert dataset, an explicitly defined network mapping for the repository’s state/control interface, trained weights, normalization metadata, and validation under the same refinement, verification, and conformal scoring pipeline. The paper also uses MINCO during multi-waypoint transitions, so it does not eliminate polynomial trajectories from every part of its full system.

No neural model, training script, or calibration data is implemented as part of this comparison.